

EXECUTIVE SUMMARY

Project Title: Instagram Engagement Analysis

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Internship: Data Science Internship – Alfido Tech

Project Objective

The primary objective of this project was to analyze Instagram engagement data and identify patterns that influence user interaction. The study focused on understanding user behavior, content performance, posting activity, and hashtag usage.

Dataset Overview

The dataset contains Instagram-related information distributed across multiple files including users, photos, likes, comments, hashtags, and follower relationships. These datasets were integrated and analyzed to generate meaningful business insights.

Data Cleaning & Preparation

The following preprocessing steps were performed:

- Checked missing values
- Inspected data types
- Converted date columns into datetime format
- Removed inconsistencies where necessary
- Created engagement metrics using likes and comments
- Merged multiple datasets for analysis

Analysis Performed

- User profile analysis
- Photo type analysis
- Engagement analysis
- Hashtag usage analysis
- Monthly posting trend analysis

Key Findings

1. Public accounts provide greater engagement opportunities than private accounts.
2. Content format plays an important role in user interaction.
3. A small number of posts generate a large proportion of total engagement.
4. Popular hashtags appear frequently among successful posts.
5. Consistent posting activity contributes to improved visibility.

Recommendations for Alfido Tech

1. Maintain a consistent content calendar.
2. Focus on content formats with higher engagement.
3. Utilize relevant and trending hashtags.

4. Encourage audience participation through interactive captions.
5. Continuously monitor performance metrics and optimize strategies.

Conclusion

The project successfully identified patterns within Instagram engagement data and provided actionable recommendations that can help Alfido Tech improve audience engagement, reach, and overall social media performance.